PinIT Career OS — 36+ Courses Master Visuals & Duration Architecture

Document Identifier	**PINIT_36_INDIVIDUAL_COURSES_VISUALS_PLAN** (v3.0 Zero-Hallucination & Duration Audit)
Total Courses & Volume	**37 Master Courses • 1,110 Lesson Days • 6,660 Interactive Lesson Parts**
Pedagogical Standard	**18–25 Minutes Lesson Duration • Deterministic Gate v2 Visual Bindings**
Actual Measured Reality	**13 Courses Already Built (30/30) • 24 Courses Standalone (Expansion + Visuals Needed)**

1. Honest Reality Audit: Already Built vs. To Be Built

An audit of the codebase confirms that some courses already have 30-day visual suites built from the Python and Web certification tracks. The 37 courses are structured into 4 reality categories:

Category	Description	Count	Visuals Status	Lesson Duration
Category A: Ready to Ship	Already built in cert tracks + LongLessons format	**12 Courses**	✅ **30/30 Built**	🎯 **21.2–24.4m (PERFECT)**
Category B: Minor Trim Needed	Visuals 30/30 built (`sql-mastery`), duration slightly high	**1 Course**	✅ **30/30 Built**	⚠️ **26.1m (Trim to 23m)**
Category C: Visuals Needed	Duration perfect (`design`), but visuals need authoring	**1 Course**	⏳ **0/30 Needed**	🎯 **22.5m (PERFECT)**
Category D: Standalone Courses	Standalone courses in short format (~6m); needs expansion & visuals	**23 Courses**	⏳ **0/30 Needed**	⚠️ **6.0m (Expand to 18-25m)**

2. The 18–25 Minute Pedagogical Duration Standard

In PinIT Career OS, lesson duration is calculated using the platform's official formula in `src/lib/data/longLessons.ts`:

> ***Total Duration*** = ***Spoken Time*** *(words ÷ 120 wpm)* + ***Hands-on Activity Time*** *(6 parts × 2.0 min per tryIt & check).*
> *Target: 1,200–1,500 words of spoken lecture (10–12.5m) + interactive coding changes and diagnostic checks (11–12m) =* ***21.0 to 24.5 minutes****.*

3. Master Course-by-Course Duration & Visuals Audit (All 37 Courses)

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
01	`course-java-logic`	**Java Fundamentals & Core Logic**	1m	5.5m	**6.5m**	**0/30**
02	`course-react-web`	**Full-Stack React Web Development**	9.5m	11.9m	**21.4m**	**30/30**
03	`course-node-web`	**Node.js & TypeScript Backend Engineering**	11.6m	12m	**23.6m**	**30/30**
04	`course-cloud-native`	**Cloud Native Architectures (AWS)**	12.1m	12m	**24.1m**	**30/30**
05	`course-devops-cicd`	**DevOps & CI/CD Pipeline Automation**	10.6m	12m	**22.6m**	**30/30**
06	`course-design-systems`	**UI/UX Design Systems & Visual Frontend**	10.5m	12m	**22.5m**	**0/30**
07	`course-dsa-optim`	**Data Structures & Algorithmic Optimizations**	11.3m	12m	**23.3m**	**30/30**
08	`course-mobile-dev`	**Mobile Application Development**	1m	5m	**6m**	**0/30**

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
09	`course-cybersecurity`	**Cybersecurity Principles & Secure Systems**	12.4m	12m	**24.4m**	**30/30**
10	`course-database-eng`	**Database Engineering & Query Performance**	14.1m	12m	**26.1m**	**30/30**
11	`course-distributed-sys`	**High-Scale Distributed System Design**	10.8m	12m	**22.8m**	**30/30**
12	`course-ai-eng`	**AI Engineering & LLM Integration**	10.6m	12m	**22.6m**	**30/30**
13	`course-fullstack-js`	**Full-Stack JavaScript Engineering**	1m	5m	**6m**	**0/30**
14	`course-iot-embedded`	**IoT, Firmware & Embedded Systems**	1m	5m	**6m**	**0/30**
15	`course-3d-graphics`	**3D Interactive Graphics & Avatar Animation**	1m	5m	**6m**	**0/30**
16	`course-blockchain-web3`	**Blockchain, Web3 & Smart Contracts**	1m	5m	**6m**	**0/30**
17	`course-iot-network`	**IoT Wireless Networks & Protocols**	1m	5m	**6m**	**0/30**
18	`course-iot-edge-ai`	**Edge AI, DSP & TinyML Systems**	1m	5m	**6m**	**0/30**
19	`course-iot-security`	**Industrial IoT Security & Device Lifecycle**	1m	5m	**6m**	**0/30**
20	`course-python-backend`	**Python Programming & Backend Systems**	10.6m	12m	**22.6m**	**30/30**
21	`course-quant-systems`	**Quantitative Engineering & Low-Latency Trading Systems**	9.4m	12m	**21.4m**	**30/30**
22	`course-digital-accounting`	**Digital Accounting & Taxation (B.Com / BBA)**	1m	5m	**6m**	**0/30**
23	`course-finance-investment`	**Business Finance & Investment Management (B.Com / BBA)**	1m	5m	**6m**	**0/30**
24	`course-business-analytics`	**Business Analytics & Decision Intelligence (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
25	`course-marketing-branding`	**Marketing & Brand Management (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
26	`course-digital-marketing`	**Digital Marketing & Growth Strategy (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
27	`course-ecommerce-digital-biz`	**E-Commerce & Digital Business (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
28	`course-entrepreneurship-biz-mgmt`	**Entrepreneurship & Business Management (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
29	`course-sales-crm-success`	**Sales, Customer Success & CRM (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
30	`course-operations-supplychain-compliance`	**Operations, Supply Chain & Business Compliance (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
31	`course-ai-digital-transformation`	**AI & Digital Transformation for Business (B.Com / BBA / MBA)**	1m	5m	**6m**	**0/30**
32	`course-computer-fundamentals`	**Computer Literacy, Digital Productivity & OS Fundamentals**	1m	5m	**6m**	**0/30**

#	Course Identifier	Title	Spoken	Activity	Total	Visuals
33	`course-ai-prompt-literacy`	**Everyday AI Literacy & Prompt Engineering**	9.2m	12m	**21.2m**	**30/30**
34	`course-excel-data-viz`	**Excel & Data Analysis Fundamentals**	1m	5m	**6m**	**0/30**
35	`course-git-version-control`	**Git, GitHub & Version Control Basics**	1m	5m	**6m**	**0/30**
36	`course-softskills-communication`	**Professional Tech Communication & Interview Mastery**	1m	5m	**6m**	**0/30**
37	`course-nlp`	**Natural Language Processing & Computational Linguistics**	9.5m	12m	**21.5m**	**30/30**

4. Master Visual Templates Catalog (17 Interactive Diagram Engines)

Template Key	Engine Name	Domain	Interactive Mechanics & Pedagogical Function
`flow`	**Multi-Stage Flow**	Universal	Linear and branching pipelines with stage status highlights
`boxes`	**Memory & State Boxes**	Systems / Web	Slots holding named variables, props, register values, or configs
`table`	**Relational & Ledger Table**	Data / Fin / API	Columns and rows with row-by-row or cell-by-cell step illumination
`letters`	**Character Array**	CompSci / NLP	Character cells with moving pointers for strings, tokens, and regex
`compare`	**Comparative Panels**	Universal	Side-by-side comparison with pass/fail indicators and pros/cons
`cells`	**1D Memory Array Buffer**	DSA / Hardware	Indexed array blocks with up to 3 named moving pointers
`stack-queue`	**LIFO Stack / FIFO Queue**	Systems / Trading	Vertical stack frames or horizontal queue with push/pop animations
`tree-graph`	**Hierarchical Tree / Graph**	DSA / Web / 3D	Up to 6 nodes with directional edges and traversal highlighting
`bars`	**Quantitative Metrics Bars**	Analytics / SRE	Labeled numeric bars comparing latency, ratios, or benchmarks
`sequence`	**Chronological Sequence**	Network / Auth	Lifeline columns with directional request/response message arrows
`states`	**Finite State Machine**	Protocols / Async	State bubbles with transitions and active state illumination
`workflow`	**Multi-Tier Topology**	Cloud / DevOps	System architecture topology connecting client, proxy, pods, DB
`component-tree`	**UI Component Hierarchy**	Frontend / Mobile	Component tree displaying props cascading down and events bubbling up
`wireframe`	**Box-Model Wireframe**	Frontend / Design	Nested UI boxes visualizing layout engines, margin, border, padding
`register-bits`	**Bitwise Register Simulator**	Embedded / IoT	8/16/32-bit register cells visualizing bitmasks, set/clear operations
`ledger-sheet`	**Double-Entry T-Account**	Commerce / Acct	T-Account debit/credit columns maintaining balancing invariants
`funnel`	**Conversion Funnel**	Marketing / Sales	Multi-stage conversion stages showing drop-off percentages and cohorts

5. Detailed Blueprint & Action Plan (All 37 Courses, One by One)

Below is the exhaustive architectural specification for every single course, specifying existing assets, measured times, allowed templates, and curriculum blocks:

Course 01: Java Fundamentals & Core Logic

`course-java-logic`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6.5 min (Expansion to 18-25m Needed) | **Allowed:** `boxes, flow, table, cells, stack-queue, compare`

Architectural Theme: JVM Architecture, Primitive Typing, OOP Invariants & Enterprise Modularity

> ***Pedagogical Whiteboard Model:*** *Illustrate JVM Stack frame pushes/pops, heap reference allocations, and variable mutations during loops. Visuals clarify value vs reference semantics.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**JVM Bytecode execution, Primitive boxing & System.out printing flow**	Template `boxes`: Animated step-by-step state progression
Block 2	Days 6–10	**Conditional branching, Relational operators & Boolean logic tables**	Template `flow`: Animated step-by-step state progression
Block 3	Days 11–15	**While/Do-While loops, Iterative counters & Array indexing cells**	Template `table`: Animated step-by-step state progression
Block 4	Days 16–20	**For loops, Nested iteration patterns & Matrix coordinate grids**	Template `cells`: Animated step-by-step state progression
Block 5	Days 21–25	**Method call frames, Parameter pass-by-value & Return value routing**	Template `stack-queue`: Animated step-by-step state progression
Block 6	Days 26–30	**OOP Class blueprints, Encapsulation & Milestone Financial Engine**	Template `compare`: Animated step-by-step state progression

Capstone Milestone: Milestone Financial Utility Engine: A multi-module Java application modeling bank accounts, compound interest calculators, and transactional debit/credit validation.

Course 02: Full-Stack React Web Development

`course-react-web`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 21.4 min (18-25m Standard) | **Allowed:** `component-tree, wireframe, flow, boxes, compare, states`

Architectural Theme: Fiber Reconciliation, Hook Closures, Virtual DOM Diffing & Component Trees

> ***Pedagogical Whiteboard Model:*** *Show component trees with props cascading downwards and callback events bubbling up. Highlight component re-renders triggered by state setter dispatches.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Declarative JSX parsing, Component composition & Virtual DOM tree**	Template `component-tree` : Animated step-by-step state progression
Block 2	Days 6–10	**useState hook closures, Immutable state updates & Re-render triggers**	Template `wireframe` : Animated step-by-step state progression
Block 3	Days 11–15	**Props contract interfaces, Prop drilling vs Context & Children nodes**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**useEffect dependency arrays, Async fetch cycles & Cleanup timers**	Template `boxes` : Animated step-by-step state progression
Block 5	Days 21–25	**Controlled forms, Validation state machines & UI layout wireframes**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Next.js App Router hybrid rendering, Server vs Client components**	Template `states` : Animated step-by-step state progression

Capstone Milestone: Production Enterprise SaaS Portal: High-performance Next.js application with dynamic dashboard routing, dark/light theme tokens, and live analytics telemetry.

Course 03: Node.js & TypeScript Backend Engineering

`course-node-web`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 23.6 min (18-25m Standard) | **Allowed:** `sequence, stack-queue, flow, boxes, table, states`

Architectural Theme: Asynchronous Event Loop, Middleware Pipelines & Relational Microservices

> ***Pedagogical Whiteboard Model:*** *Map incoming HTTP request flows through Express middleware waterfalls. Show database connection pool acquisitions and async non-blocking I/O queues.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**V8 Engine Call Stack, libuv Event Loop & Microtask/Macrotask queues**	Template `sequence` : Animated step-by-step state progression
Block 2	Days 6–10	**TypeScript strict typing, Interface schemas & Compile-time validation**	Template `stack-queue` : Animated step-by-step state progression
Block 3	Days 11–15	**HTTP Request-Response lifecycles, RESTful routes & Express middleware**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Relational PostgreSQL queries, Connection pooling & ACID transactions**	Template `boxes` : Animated step-by-step state progression
Block 5	Days 21–25	**JWT HMAC-SHA256 authentication, Bcrypt password hashing & Auth guards**	Template `table` : Animated step-by-step state progression
Block 6	Days 26–30	**Containerized production Docker build & Health-check endpoint telemetry**	Template `states` : Animated step-by-step state progression

Capstone Milestone: High-Concurrency REST Microservice: Authenticated backend API with role-based access control, connection-pooled PostgreSQL persistence, and Docker multi-stage deployment.

Course 04: Cloud Native Architectures (AWS)

`course-cloud-native`

Visuals: ✅ **Built (30/30)** | **Duration:** 🎯 **24.1 min (18-25m Standard)** | **Allowed:** `workflow, sequence, states, table, bars, compare`

Architectural Theme: AWS VPC Topologies, Serverless Lifecycles & Highly Available Cloud Systems

> ***Pedagogical Whiteboard Model:*** *Visualize cloud architecture topologies: VPC subnets, Internet Gateways, Application Load Balancers, and ECS tasks. Clarify IAM role evaluation logic.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**AWS Global Cloud Infrastructure, Regions, Availability Zones & IAM Policies**	Template `workflow` : Animated step-by-step state progression
Block 2	Days 6–10	**VPC CIDR subnets, Public vs Private routing & Security Group firewalls**	Template `sequence` : Animated step-by-step state progression
Block 3	Days 11–15	**Compute scaling: EC2 instances, Application Load Balancers & Auto Scaling**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**Serverless computing: AWS Lambda triggers, API Gateway & DynamoDB NoSQL**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Container orchestration: ECR image registries & ECS/Fargate task definitions**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Multi-AZ High Availability, Route53 DNS failover & Terraform IaC blueprints**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Multi-AZ Serverless Cloud Infrastructure: Terraform-provisioned VPC architecture with ALB traffic distribution, ECS Fargate microservices, and automated health checks.

Course 05: DevOps & CI/CD Pipeline Automation

`course-devops-cicd`

Visuals: ✅ **Built (30/30)** | **Duration:** 🎯 **22.6 min (18-25m Standard)** | **Allowed:** `flow, workflow, states, sequence, compare`

Architectural Theme: Immutable Containerization, Automated CI/CD Pipelines & Kubernetes Orchestration

> ***Pedagogical Whiteboard Model:*** *Diagram Docker layer caching trees, GitHub Actions stage execution waterfalls, and Kubernetes ReplicaSet pod reconciliation loops.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**POSIX Linux administration, Shell piping & Systemd daemon services**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Git trunk-based branching, Semantic versioning & GitOps workflows**	Template `workflow` : Animated step-by-step state progression
Block 3	Days 11–15	**Docker engine internals, Multi-stage image builds & Layer caching DAGs**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**GitHub Actions runners, Lint-Test-Build matrix workflows & Secret vaults**	Template `sequence` : Animated step-by-step state progression
Block 5	Days 21–25	**Kubernetes cluster architecture: Control plane, Pods, Services & Ingress**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Zero-downtime Blue/Green deployments, Canary rollouts & Prometheus alerts**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Enterprise Automated GitOps Pipeline: Multi-stage Docker containerization pipeline with automated GitHub Actions testing, security vulnerability scanning, and Kubernetes deployment.

Course 06: UI/UX Design Systems & Visual Frontend

`course-design-systems`

Visuals: ⏳ Needed (0/30) | **Duration:** 🎯 22.5 min (18-25m Standard) | **Allowed:** `wireframe, component-tree, compare, boxes, bars`

Architectural Theme: Atomic Component Architectures, Design Tokens & Accessible Web Layouts

> ***Pedagogical Whiteboard Model:*** *Draw CSS box-model margins/borders/padding, Flexbox/Grid spatial coordinates, and accessibility contrast verification ratios across light and dark modes.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Visual hierarchy principles, Modular typography scales & 8pt grid geometry**	Template `wireframe` : Animated step-by-step state progression
Block 2	Days 6–10	**Color space tokens, HSL palette generation & WCAG AAA contrast ratios**	Template `component-tree` : Animated step-by-step state progression
Block 3	Days 11–15	**Atomic Design methodology: Atoms, Molecules, Organisms & Templates**	Template `compare` : Animated step-by-step state progression
Block 4	Days 16–20	**CSS Layout mechanics: Flexbox axes, CSS Grid fr tracks & Responsive breakpoints**	Template `boxes` : Animated step-by-step state progression
Block 5	Days 21–25	**Interactive component state machines: Hover, Focus-visible, Active & Disabled**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Design Token architecture: Style Dictionary, JSON token exports & Figma sync**	Template `wireframe` : Animated step-by-step state progression

Capstone Milestone: Production Design System Library: Documented, accessible UI component kit featuring tokens, atomic components, and responsive mobile-to-desktop wireframe layouts.

Course 07: Data Structures & Algorithmic Optimizations

`course-dsa-optim`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 23.3 min (18-25m Standard) | **Allowed:** `cells, stack-queue, tree-graph, table, bars, compare`

Architectural Theme: Asymptotic Complexity, Memory Buffers, Tree Traversal & Dynamic Programming

> ***Pedagogical Whiteboard Model:*** *Show memory array buffers with dual pointers shifting, recursion stack frame winding/unwinding, and 2D memoization grids populating optimal substructure solutions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Big-O Asymptotic Space-Time analysis, Linear arrays & In-place operations**	Template `cells` : Animated step-by-step state progression
Block 2	Days 6–10	**Two-pointer convergence, Sliding window bounds & Prefix sum caching**	Template `stack-queue` : Animated step-by-step state progression
Block 3	Days 11–15	**Linked memory lists, LIFO Call Stacks & FIFO Circular Queue buffers**	Template `tree-graph` : Animated step-by-step state progression
Block 4	Days 16–20	**Hierarchical Trees, Binary Search Tree balancing & Binary Heap priorities**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Graph topologies, Adjacency lists, BFS/DFS traversal & Dijkstra shortest paths**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Dynamic Programming: Memoization tables, Tabulation & Knapsack optimization**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: High-Performance Algorithmic Optimization Suite: Benchmark-validated implementations of cache-conscious searching, graph pathfinding, and dynamic programming schedulers.

Course 08: Mobile Application Development

`course-mobile-dev`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `wireframe, component-tree, flow, sequence, states`

Architectural Theme: Cross-Platform React Native, JSI Runtime, Native Modules & Offline SQLite

> ***Pedagogical Whiteboard Model:*** *Illustrate JavaScript-to-Native bridge execution, Yoga Flexbox layout reflows, and touch gesture responder state transitions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**React Native architecture: JavaScript thread, JSI, Shadow Tree & Native UI**	Template `wireframe` : Animated step-by-step state progression
Block 2	Days 6–10	**Core UI primitives: View, Text, FlatList virtualization & Yoga flex layouts**	Template `component-tree` : Animated step-by-step state progression
Block 3	Days 11–15	**Navigation architectures: Native Stack, Bottom Tabs & Deep-link URI routing**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Offline state persistence: MMKV fast key-value & SQLite local transactions**	Template `sequence` : Animated step-by-step state progression
Block 5	Days 21–25	**Native hardware bridges: Camera capture, GPS Geolocation & Biometric auth**	Template `states` : Animated step-by-step state progression
Block 6	Days 26–30	**Performance profiling: Hermes bytecode compilation & Release bundle packaging**	Template `wireframe` : Animated step-by-step state progression

Capstone Milestone: Cross-Platform Mobile Utility App: Feature-complete React Native application featuring virtualized infinite feeds, offline SQLite synchronization, and biometric authentication.

Course 09: Cybersecurity Principles & Secure Systems

`course-cybersecurity`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 24.4 min (18-25m Standard) | **Allowed:** `flow, sequence, table, compare, states`

Architectural Theme: Cryptographic Protocols, OWASP Top 10 Mitigation & Zero-Trust Defense

> ***Pedagogical Whiteboard Model:*** *Diagram packet inspection sequences, TLS 1.3 cryptographic key exchanges, JWT tampering detection, and SQL injection sanitization barriers.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Threat modeling methodologies, Attack vectors & OWASP Top 10 vulnerabilities**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Authentication architecture: Multi-factor auth, Salted Bcrypt & Session stores**	Template `sequence` : Animated step-by-step state progression
Block 3	Days 11–15	**Cryptographic systems: Symmetric AES-GCM, Asymmetric RSA & ECC key pairs**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Web application hardening: CSP headers, CORS policies & CSRF token validation**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Network defense: TLS 1.3 handshake, WAF firewall rules & DDoS rate limiters**	Template `states` : Animated step-by-step state progression
Block 6	Days 26–30	**Security incident response: Penetration testing workflows & SIEM audit logs**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Zero-Trust Secure Authentication Gateway: End-to-end hardened authentication microservice with Argon2id password hashing, rotating JWT signing keys, and brute-force rate limiting.

Course 10: Database Engineering & Query Performance

`course-database-eng`

Visuals: ✅ Built (30/30) | **Duration:** ⚠️ 26.1 min (Expansion to 18-25m Needed) | **Allowed:** `table, tree-graph, flow, compare, bars`

Architectural Theme: Relational Theory, B+ Tree Indexes, Transaction Isolation & WAL Logging

> ***Pedagogical Whiteboard Model:*** *Illustrate B+ Tree node splits, Write-Ahead Log (WAL) sequential flushes, and query execution plans comparing sequential table scans with index lookups.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Relational algebra, Entity-Relationship schemas & Normalization (1NF–BCNF)**	Template `table` : Animated step-by-step state progression
Block 2	Days 6–10	**SQL Query execution engine: Parsing, Query planning & AST optimization**	Template `tree-graph` : Animated step-by-step state progression
Block 3	Days 11–15	**Indexing data structures: B+ Tree page balancing, Hash indexes & Covering indexes**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**ACID properties, MVCC concurrency control & Transaction isolation anomalies**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Query profiling: EXPLAIN ANALYZE interpretation & Buffer cache hit ratios**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Horizontal scaling: Table partitioning, Read replication & Connection pools**	Template `table` : Animated step-by-step state progression

Capstone Milestone: High-Throughput Financial Ledger Database: Production-tuned PostgreSQL schema with B+ tree composite indexes, strict MVCC isolation, and sub-millisecond query execution plans.

Course 11: High-Scale Distributed System Design

`course-distributed-sys`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 22.8 min (18-25m Standard) | **Allowed:** `sequence, workflow, states, table, cells, compare`

Architectural Theme: Consensus Protocols, Consistent Hashing, Distributed State & Fault Tolerance

> ***Pedagogical Whiteboard Model:*** *Show consistent hashing token rings, Raft leader election heartbeat sequences, and 2-Phase Commit prepare/commit transaction message flows.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Fallacies of distributed computing, Network partitions & CAP/PACELC theorems**	Template `sequence` : Animated step-by-step state progression
Block 2	Days 6–10	**RPC protocols, Protocol Buffers schema compilation & gRPC streaming channels**	Template `workflow` : Animated step-by-step state progression
Block 3	Days 11–15	**Data partitioning: Consistent hashing rings, Virtual nodes & Key distribution**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**Consensus algorithms: Paxos & Raft leader election, Term leases & Log commits**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Distributed transactions: Two-Phase Commit (2PC) & Saga compensating patterns**	Template `cells` : Animated step-by-step state progression
Block 6	Days 26–30	**Resilience patterns: Circuit breakers, Distributed rate limiting & Jaeger tracing**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Fault-Tolerant Distributed Key-Value Store: Raft-replicated cluster featuring automated leader failover, consistent hashing shard allocation, and vector clock conflict detection.

Course 12: AI Engineering & LLM Integration

`course-ai-eng`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 22.6 min (18-25m Standard) | **Allowed:** `flow, table, bars, sequence, compare, boxes`

Architectural Theme: Large Language Model Systems, Vector Similarity Search, RAG & Autonomous Agents

> ***Pedagogical Whiteboard Model:*** *Draw token embedding coordinate spaces, RAG vector retrieval cosine similarity rankings, and agentic tool dispatching request/response loops.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Transformer architecture fundamentals: Tokenization, Embeddings & Self-Attention**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Prompt engineering patterns: Few-shot context, System framing & Chain-of-Thought**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Vector databases: Dense embeddings, HNSW indexing & Cosine similarity search**	Template `bars` : Animated step-by-step state progression
Block 4	Days 16–20	**Retrieval-Augmented Generation (RAG): Document chunking & Context injection**	Template `sequence` : Animated step-by-step state progression
Block 5	Days 21–25	**Agentic tool calling: JSON function signatures, API dispatch & ReAct loops**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Evaluation & Guardrails: Hallucination detection, Token budgets & Model routing**	Template `boxes` : Animated step-by-step state progression

Capstone Milestone: Enterprise Document Intelligence Copilot: Production RAG pipeline with semantic document chunking, hybrid vector search, citation synthesis, and guardrail verification.

Course 13: Full-Stack JavaScript Engineering

`course-fullstack-js`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, component-tree, sequence, table, states, compare`

Architectural Theme: Monolithic to Microservice JS, Next.js Server Components, ORMs & WebSockets

> ***Pedagogical Whiteboard Model:*** *Show Next.js Server Component streaming hydration boundaries, WebSocket bidirectional frame ping-pong, and database mutation waterfalls.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Modern ESNext language features, Microtasks, Async/Await & Generator functions**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Next.js 14 App Router: Server Components, Streaming Suspense & Route Handlers**	Template `component-tree` : Animated step-by-step state progression
Block 3	Days 11–15	**Server Actions, Client component boundaries & Optimistic UI updates**	Template `sequence` : Animated step-by-step state progression
Block 4	Days 16–20	**Type-safe ORM persistence: Prisma schemas, Relational joins & DB migrations**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Real-time communication: WebSocket duplex channels & Server-Sent Events (SSE)**	Template `states` : Animated step-by-step state progression
Block 6	Days 26–30	**Production deployment: Edge middleware, Vercel serverless & Dockerized images**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Real-Time Collaborative Workspace: Full-stack application with live multi-user WebSocket editing, Prisma database synchronization, and optimistic UI transitions.

Course 14: IoT, Firmware & Embedded Systems

`course-iot-embedded`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, states, cells, boxes, sequence, register-bits`

Architectural Theme: Hardware Registers, Memory-Mapped I/O, Interrupts & FreeRTOS Schedulers

> ***Pedagogical Whiteboard Model:*** *Diagram microcontroller memory-mapped registers, bitwise mask operations, interrupt vector dispatching, and FreeRTOS task priority preemption.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**MCU architectures (ARM Cortex-M), Memory map & GPIO register addressing**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Bitwise register manipulation: Bitmasking, Atomic Set/Reset (BSRR) & Pull-ups**	Template `states` : Animated step-by-step state progression
Block 3	Days 11–15	**Timers, PWM signal generation & Hardware Interrupt Service Routines (ISR)**	Template `cells` : Animated step-by-step state progression
Block 4	Days 16–20	**Analog-to-Digital Conversion (ADC): Quantization step sizing & Two-point scaling**	Template `boxes` : Animated step-by-step state progression
Block 5	Days 21–25	**Serial protocols: UART baud rates, I2C 7-bit bus addressing & SPI clock polarity**	Template `sequence` : Animated step-by-step state progression
Block 6	Days 26–30	**Real-Time Operating Systems: FreeRTOS preemptive scheduler, Queues & Semaphores**	Template `register-bits` : Animated step-by-step state progression

Capstone Milestone: Embedded Industrial Telemetry Controller: FreeRTOS firmware implementing multi-channel ADC sensor acquisition, lock-free ring buffers, and I2C peripheral telemetry.

Course 15: 3D Interactive Graphics & Avatar Animation

`course-3d-graphics`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `tree-graph, boxes, flow, compare, bars`

Architectural Theme: Vector Math, WebGL Render Pipeline, Shaders, Scene Graphs & Skeletal Rigging

> ***Pedagogical Whiteboard Model:*** *Visualize the 3D Model-View-Projection (MVP) matrix transformation pipeline, shader rasterization stages, and hierarchical skeletal bone transforms.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**3D Coordinate geometry: Cartesian vectors, Dot/Cross products & Matrix transforms**	Template `tree-graph` : Animated step-by-step state progression
Block 2	Days 6–10	**WebGL rendering pipeline: Vertex Shaders, Fragment Shaders & Rasterization**	Template `boxes` : Animated step-by-step state progression
Block 3	Days 11–15	**Three.js scene graph architecture: Object3D hierarchy, Meshes & Geometry buffers**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Material physics & Lighting: Ambient, Point, Directional & Physically Based Rendering**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Skeletal animation: Rigged bone hierarchies, Skinning weights & Keyframe tracks**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Rendering optimizations: Draw call batching, Frustum culling & LOD geometries**	Template `tree-graph` : Animated step-by-step state progression

Capstone Milestone: Interactive 3D Avatar Animation Studio: WebGL-accelerated 3D viewport featuring custom PBR lighting shaders, skeletal bone transforms, and interactive orbital controls.

Course 16: Blockchain, Web3 & Smart Contracts

`course-blockchain-web3`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `tree-graph, stack-queue, flow, sequence, compare, states`

Architectural Theme: Cryptographic Hashing, Merkle Trees, EVM Gas Mechanics & Smart Contracts

> ***Pedagogical Whiteboard Model:*** *Show Merkle tree cryptographic proof verification, EVM memory/stack gas consumption charts, and reentrancy attack state transition barriers.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Cryptographic hash functions (SHA-256, Keccak), Asymmetric signatures & P2P networks**	Template `tree-graph` : Animated step-by-step state progression
Block 2	Days 6–10	**Blockchain ledger structure: Block headers, Merkle Patricia Tries & Consensus**	Template `stack-queue` : Animated step-by-step state progression
Block 3	Days 11–15	**Ethereum Virtual Machine (EVM): Execution stack, Storage slots & Gas accounting**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Solidity contract development: Value types, Access modifiers & Event emissions**	Template `sequence` : Animated step-by-step state progression
Block 5	Days 21–25	**Smart contract security: Checks-Effects-Interactions, Reentrancy & Integer safety**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Decentralized frontend: Ethers.js / Viem, Metamask wallet RPC & Token transactions**	Template `states` : Animated step-by-step state progression

Capstone Milestone: Decentralized Escrow Protocol & dApp: Production Solidity smart contract with multi-signature release conditions, reentrancy guards, and reactive Web3 frontend.

Course 17: IoT Wireless Networks & Protocols

`course-iot-network`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, sequence, states, table, compare`

Architectural Theme: Wireless RF Topologies, BLE GATT Profiles, LoRaWAN Chirps & MQTT Brokers

> ***Pedagogical Whiteboard Model:*** *Map BLE GATT Service/Characteristic hierarchies, LoRaWAN chirp spread spectrum uplinks, and MQTT publish/subscribe broker message routing.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Wireless RF propagation fundamentals, Frequencies (ISM bands) & Network topologies**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Bluetooth Low Energy (BLE): GAP advertising, GATT Services & Characteristics**	Template `sequence` : Animated step-by-step state progression
Block 3	Days 11–15	**IEEE 802.15.4 & Zigbee Mesh: Multi-hop routing & Self-healing network trees**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**LoRa & LoRaWAN: Chirp Spread Spectrum (CSS), Gateway uplinks & Join procedures**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**IoT Application protocols: MQTT QoS 0/1/2 handshakes, CoAP & Lightweight M2M**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Ultra-low-power optimization: Deep sleep modes, Duty cycling & Energy budgets**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Low-Power LoRaWAN Environmental Gateway: Edge mesh network firmware routing environmental sensor telemetry through encrypted LoRaWAN packets to an MQTT cloud broker.

Course 18: Edge AI, DSP & TinyML Systems

`course-iot-edge-ai`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, bars, cells, table, compare`

Architectural Theme: Digital Signal Processing, Feature Extraction, INT8 Quantization & TinyML

> ***Pedagogical Whiteboard Model:*** *Show sliding window sensor buffers, Fast Fourier Transform (FFT) spectrograms, and INT8 quantized weight matrices mapped to MCU memory arenas.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Edge intelligence paradigms, Microcontroller resource budgets (SRAM/Flash)**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Digital Signal Processing (DSP): Discrete filtering, Windowing & Fast Fourier Transform**	Template `bars` : Animated step-by-step state progression
Block 3	Days 11–15	**Time-series feature extraction: Spectral power, Zero-crossing rates & Peak detection**	Template `cells` : Animated step-by-step state progression
Block 4	Days 16–20	**Model optimization for edge: Post-training INT8 quantization & Weight pruning**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**TensorFlow Lite for Microcontrollers (TFLM): Memory arenas & Flatbuffer models**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Real-time inference pipeline: Accelerometer gesture recognition & Anomaly detection**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Edge TinyML Gesture Recognition Engine: Microcontroller-deployed model classifying continuous IMU sensor streams in real-time under a 32 KB SRAM memory constraint.

Course 19: Industrial IoT Security & Device Lifecycle

`course-iot-security`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, sequence, states, compare, table`

Architectural Theme: Hardware Root of Trust, Secure Boot, Mutual TLS & Secure OTA Lifecycles

> ***Pedagogical Whiteboard Model:*** *Diagram secure boot cryptographic signature verification chains, mTLS certificate exchanges, and dual-bank OTA firmware flash memory rollback safety.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Industrial IoT threat vectors, Purdue Enterprise Reference Model & Attack surfaces**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Hardware Root of Trust: Secure Elements (ATECC608), TPMs & Cryptographic coprocessors**	Template `sequence` : Animated step-by-step state progression
Block 3	Days 11–15	**Secure Boot architectures: Cryptographic hash validation & Anti-rollback eFuses**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**Identity & Transport security: X.509 Device certificates & Mutual TLS (mTLS)**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Secure Over-The-Air (OTA) updates: Dual-bank flash partitions & Rollback protection**	Template `table` : Animated step-by-step state progression
Block 6	Days 26–30	**Industrial communication security: Encrypted Modbus-TCP, OPC UA & IEC 62443**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Hardened Industrial Edge Controller: Secure boot-verified firmware implementing mutual TLS cloud authentication and failsafe dual-bank OTA update recovery.

Course 20: Python Programming & Backend Systems

`course-python-backend`

Visuals: ✅ **Built (30/30)** | **Duration:** 🎯 **22.6 min (18-25m Standard)** | **Allowed:** `flow, sequence, boxes, table, states, compare`

Architectural Theme: Python Asynchronous Systems, FastAPI, Pydantic & Distributed Task Queues

> ***Pedagogical Whiteboard Model:*** *Illustrate asyncio event loop coroutine yields, Pydantic JSON schema validations, and Celery asynchronous task distribution across Redis workers.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Advanced Python paradigms: Context managers, Decorators & Generator iterables**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Asynchronous programming: asyncio event loops, Coroutine tasks & Non-blocking I/O**	Template `sequence` : Animated step-by-step state progression
Block 3	Days 11–15	**FastAPI architecture: Path operations, Pydantic type models & Dependency Injection**	Template `boxes` : Animated step-by-step state progression
Block 4	Days 16–20	**Database persistence: SQLAlchemy 2.0 Async Session, PostgreSQL & Alembic migrations**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Distributed task execution: Celery workers, Redis message brokers & Background tasks**	Template `states` : Animated step-by-step state progression
Block 6	Days 26–30	**Production engineering: Docker containerization, Pytest unit tests & Rate limiters**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: High-Performance Asynchronous FastAPI Service: Production microservice featuring async database connection pooling, background Celery processing, and OpenAPI contracts.

Course 21: Quantitative Engineering & Low-Latency Trading Systems

`course-quant-systems`

Visuals: ✅ **Built (30/30)** | **Duration:** 🎯 **21.4 min (18-25m Standard)** | **Allowed:** `stack-queue, table, bars, flow, cells, compare`

Architectural Theme: Limit Order Books, VWAP Slicing, Low-Latency Sockets & Microsecond Systems

> ***Pedagogical Whiteboard Model:*** *Show Limit Order Book (LOB) bid/ask price queues, execution algorithms slicing volume over time, and kernel-bypass zero-copy socket buffer flows.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Market microstructure: Limit Order Books (LOB), Depth of market & Bid-Ask spreads**	Template `stack-queue` : Animated step-by-step state progression
Block 2	Days 6–10	**Financial timeseries analytics: Returns, Rolling volatility & Log price transforms**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Algorithmic execution strategies: VWAP, TWAP, Implementation shortfall & Slippage**	Template `bars` : Animated step-by-step state progression
Block 4	Days 16–20	**Low-latency system engineering: Zero-copy buffers, Cache locality & Kernel bypass**	Template `flow` : Animated step-by-step state progression
Block 5	Days 21–25	**Risk modeling & Portfolio metrics: Value at Risk (VaR), Sharpe Ratio & Max Drawdown**	Template `cells` : Animated step-by-step state progression
Block 6	Days 26–30	**High-frequency backtesting engine: Event-driven matching simulation & FIX protocol**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Microsecond Limit Order Book Matching Engine: High-frequency order matching engine simulating sub-millisecond price-time priority execution and real-time PnL tracking.

Course 22: Digital Accounting & Taxation (B.Com / BBA)

`course-digital-accounting`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `table, flow, compare, bars, ledger-sheet`

Architectural Theme: Double-Entry Invariants, Ledger Balancing, GST Taxation & ERP Compliance

> ***Pedagogical Whiteboard Model:*** *Draw double-entry debit/credit ledger columns maintaining balancing equality, 3-way invoice matching flows, and GST tax credit cascading waterfalls.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Accounting equation fundamentals: Assets = Liabilities + Equity & Business Entity rule**	Template `table` : Animated step-by-step state progression
Block 2	Days 6–10	**Golden rules of accounting: Real, Personal & Nominal accounts & Journal posting**	Template `flow` : Animated step-by-step state progression
Block 3	Days 11–15	**Ledger posting: T-Accounts balancing, Closing balances & Trial Balance checksums**	Template `compare` : Animated step-by-step state progression
Block 4	Days 16–20	**Financial reporting: Trading Account, Profit & Loss statement & Balance Sheet layout**	Template `bars` : Animated step-by-step state progression
Block 5	Days 21–25	**Taxation architecture: GST Input Tax Credit cascading & TDS statutory deductions**	Template `ledger-sheet` : Animated step-by-step state progression
Block 6	Days 26–30	**Modern ERP workflows: Automated bank reconciliations & Suspense account resolution**	Template `table` : Animated step-by-step state progression

Capstone Milestone: Automated Corporate Accounting & GST Audit Engine: End-to-end digital accounting ledger with automated journal entries, balancing trial balances, and GST compliance validation.

Course 23: Business Finance & Investment Management (B.Com / BBA)

`course-finance-investment`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `table, bars, flow, compare`

Architectural Theme: Time Value of Money, DCF Valuation, WACC Capital Structures & Portfolio Risk

> ***Pedagogical Whiteboard Model:*** *Visualize Discounted Cash Flow (DCF) timelines, DuPont 3-stage ROE breakdowns, and debt vs equity cost weights in WACC capital structure models.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Time Value of Money (TVM): Present Value, Future Value & Compounding schedules**	Template `table` : Animated step-by-step state progression
Block 2	Days 6–10	**Capital budgeting evaluation: Net Present Value (NPV), IRR & Payback period hurdles**	Template `bars` : Animated step-by-step state progression
Block 3	Days 11–15	**Cost of Capital: Weighted Average Cost of Capital (WACC) & Optimal Debt-Equity mix**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Working capital dynamics: Cash Conversion Cycle, Operating cycle & Liquidity ratios**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Corporate Valuation: DCF terminal values, Multiples analysis & DuPont 3-stage ROE**	Template `table` : Animated step-by-step state progression
Block 6	Days 26–30	**Modern Portfolio Theory: Efficient frontier, Sharpe ratio & Beta risk sensitivity**	Template `bars` : Animated step-by-step state progression

Capstone Milestone: Corporate Financial Valuation & Investment Model: Comprehensive DCF valuation model with dynamic WACC sensitivity matrices, scenario planning, and working capital forecasts.

Course 24: Business Analytics & Decision Intelligence (B.Com / BBA / MBA) `course-business-analytics`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `bars, table, flow, compare`

Architectural Theme: Statistical Distributions, Pareto ABC Analysis, Regression & Executive KPIs

> ***Pedagogical Whiteboard Model:*** *Show descriptive distribution curves, Pareto 80/20 ABC classification breakdowns, and linear regression trendlines against executive scorecard KPIs.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Data-driven decision framework: Descriptive statistics, Mean, Median & Standard deviation**	Template `bars` : Animated step-by-step state progression
Block 2	Days 6–10	**Data distributions: Normal curves, Skewness, Outlier detection & Interquartile range**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Business KPI architecture: Balanced scorecards, Attribution & Performance variance**	Template `flow` : Animated step-by-step state progression
Block 4	Days 16–20	**Predictive analytics: Simple & Multiple OLS Linear regression & Trend forecasting**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Classification & Segmentation: Customer RFM analysis & Logistic regression odds**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Executive data storytelling: Automated KPI dashboards & Decision intelligence reports**	Template `table` : Animated step-by-step state progression

Capstone Milestone: Executive Business Intelligence & Decision Dashboard: End-to-end analytics report synthesizing customer segmentation, revenue regression forecasts, and automated KPI alert metrics.

Course 25: Marketing & Brand Management (B.Com / BBA / MBA) `course-marketing-branding`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, compare, table, bars`

Architectural Theme: Strategic Frameworks, Consumer Psychology, Brand Equity & PLC Lifecycles

> ***Pedagogical Whiteboard Model:*** *Map customer journey stages, Keller brand equity pyramid levels, BCG growth-share matrix quadrants, and omnichannel campaign ROI bar charts.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Strategic marketing foundations: 4Ps Marketing Mix, STP & Porter's Five Forces**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Consumer psychology: Cognitive decision journey, Perception biases & Social proof**	Template `compare` : Animated step-by-step state progression
Block 3	Days 11–15	**Brand identity architecture: Brand personality, Archetypes & Keller's Brand Equity**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Product portfolio management: Boston Consulting Group (BCG) Matrix & Product Life Cycle**	Template `bars` : Animated step-by-step state progression
Block 5	Days 21–25	**Integrated Marketing Communications (IMC): Omnichannel messaging & Content strategies**	Template `flow` : Animated step-by-step state progression
Block 6	Days 26–30	**Marketing performance measurement: Customer Lifetime Value (CLV) & Brand sentiment**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Comprehensive Brand Strategy & Go-To-Market Blueprint: Multi-channel brand positioning framework complete with customer persona journeys, messaging matrices, and ROI forecasts.

Course 26: Digital Marketing & Growth Strategy (B.Com / BBA / MBA) `course-digital-marketing`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, bars, table, compare, funnel`

Architectural Theme: Acquisition Funnels, Ad Auction Rankings, SEO Architectures & CAC/LTV Unit Economics

> ***Pedagogical Whiteboard Model:*** *Diagram conversion funnel drop-off waterfalls, ad auction rank bid vs quality score calculations, and CAC payback period bar comparisons.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Digital acquisition landscape: Inbound vs Outbound & Customer acquisition funnels**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Search Engine Optimization (SEO): Crawl budgets, Technical SEO, On-page & Backlinks**	Template `bars` : Animated step-by-step state progression
Block 3	Days 11–15	**Search Engine Marketing (SEM): Google Ads auction mechanics, Quality Score & CPC bidding**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Paid Social Advertising: Meta Ad pixel tracking, Lookalike audiences & ROAS targets**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Email marketing automation: Behavioral drip sequences, Open rate & Churn telemetry**	Template `funnel` : Animated step-by-step state progression
Block 6	Days 26–30	**Growth analytics: Multi-touch attribution models, A/B testing & CAC-to-LTV payback**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Growth Marketing & Performance Campaign Architecture: Complete multi-channel digital acquisition campaign with SEO keyword architecture, ad auction bidding models, and CAC/LTV funnels.

Course 27: E-Commerce & Digital Business (B.Com / BBA / MBA) `course-ecommerce-digital-biz`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, table, sequence, bars, compare, funnel`

Architectural Theme: SKU Variant Matrices, Volumetric Logistics, Checkout Gateways & Marketplace Scaling

> ***Pedagogical Whiteboard Model:*** *Illustrate e-commerce checkout conversion funnels, payment gateway authorization handshakes, and dimensional weight shipping cost calculations.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**E-Commerce business models: Direct-to-Consumer (D2C), B2B & Multi-vendor marketplaces**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Product catalog engineering: Parent-child SKU variants, Attributes & Inventory sync**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Checkout optimization: Cart abandonment mitigation, Payment gateways & Webhooks**	Template `sequence` : Animated step-by-step state progression
Block 4	Days 16–20	**Supply chain fulfillment: 3PL logistics, Dimensional weight & Last-mile delivery**	Template `bars` : Animated step-by-step state progression
Block 5	Days 21–25	**Marketplace scaling: Amazon Buy Box algorithms, Seller metrics & Commission fees**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Cross-border digital commerce: Currency localization, Duties, Taxes & Compliance**	Template `funnel` : Animated step-by-step state progression

Capstone Milestone: Scalable E-Commerce Operations Blueprint: Full operational framework for high-volume digital commerce, including SKU catalog architecture, checkout conversion funnels, and logistics routing.

Course 28: Entrepreneurship & Business Management (B.Com / BBA / MBA)

`course-entrepreneurship-biz-mgmt`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, table, compare, bars`

Architectural Theme: Business Model Canvas, TAM/SAM/SOM Sizing, Cap Table Dilution & Runway Dynamics

> ***Pedagogical Whiteboard Model:*** *Show Business Model Canvas component linkages, bottom-up TAM/SAM/SOM concentric circles, equity dilution round-by-round cap tables, and monthly burn rate runway curves.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Entrepreneurial mindset, Problem-Solution fit & Lean Startup iterative hypothesis**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Business Model Canvas (BMC): Revenue streams, Cost structures & Value propositions**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Market opportunity sizing: Total Addressable Market (TAM, SAM, SOM) calculation**	Template `compare` : Animated step-by-step state progression
Block 4	Days 16–20	**Startup financial modeling: Unit economics, Burn rate & Cash runway forecasting**	Template `bars` : Animated step-by-step state progression
Block 5	Days 21–25	**Venture financing mechanics: Term sheets, Convertible notes, SAFEs & Cap table dilution**	Template `flow` : Animated step-by-step state progression
Block 6	Days 26–30	**Operational scaling: Hiring frameworks, Agile organizational design & Milestone gates**	Template `table` : Animated step-by-step state progression

Capstone Milestone: Venture Creation & Investor-Ready Business Plan: Comprehensive startup plan featuring validated Business Model Canvas, bottom-up market sizing, unit economics, and 5-year pro forma financials.

Course 29: Sales, Customer Success & CRM (B.Com / BBA / MBA)

`course-sales-crm-success`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, table, states, bars, compare, funnel`

Architectural Theme: BANT/MEDDIC Qualification, Sales Velocity, CRM Workflows & Customer Retention

> ***Pedagogical Whiteboard Model:*** *Diagram sales velocity formula components, MEDDIC qualification scorecards, lead-to-opportunity state progression, and customer health score distributions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Modern B2B sales methodologies: Consultative selling, Challenger sales & Inbound leads**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Opportunity qualification frameworks: BANT (Budget, Authority, Need, Timing) & MEDDIC**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Sales pipeline mechanics: Deal stages, Probability weightings & Sales Velocity formula**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**CRM systems architecture: Lead routing workflows, Activity tracking & Pipeline automation**	Template `bars` : Animated step-by-step state progression
Block 5	Days 21–25	**Commercial negotiation: Objection handling frameworks, Value selling & Closing tactics**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Customer Success & Retention: Net Retention Rate (NRR), Churn telemetry & Health scores**	Template `funnel` : Animated step-by-step state progression

Capstone Milestone: Enterprise B2B Sales & CRM Operational Playbook: Configured CRM pipeline architecture with automated MEDDIC qualification scoring, sales velocity tracking, and retention playbooks.

Course 30: Operations, Supply Chain & Business Compliance (B.Com / BBA / MBA) `course-operations-supplychain-compliance`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, table, bars, compare`

Architectural Theme: SIPOC Value Stream Mapping, EOQ Inventory Models, Kraljic Matrices & Quality Control

> ***Pedagogical Whiteboard Model:*** *Draw SIPOC value stream process flows, Economic Order Quantity (EOQ) cost trade-off curves, Kraljic matrix supplier quadrants, and Six Sigma defect distributions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Operations strategy: Process architecture, Capacity planning & Bottleneck identification**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Value Stream Mapping: SIPOC diagrams, Lead time reduction & Waste elimination (Muda)**	Template `table` : Animated step-by-step state progression
Block 3	Days 11–15	**Inventory optimization: Economic Order Quantity (EOQ), Safety stock & Reorder points**	Template `bars` : Animated step-by-step state progression
Block 4	Days 16–20	**Strategic procurement: Kraljic purchasing portfolio matrix & Vendor SLA evaluations**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Total Quality Management (TQM): Six Sigma DMAIC cycle, Statistical process control & Kaizen**	Template `flow` : Animated step-by-step state progression
Block 6	Days 26–30	**Business compliance: Supply chain ethics, ISO certifications & Regulatory audit governance**	Template `table` : Animated step-by-step state progression

Capstone Milestone: End-to-End Supply Chain Optimization Plan: Operations master plan featuring value stream mapping, automated EOQ inventory replenishment models, and supplier quality audits.

Course 31: AI & Digital Transformation for Business (B.Com / BBA / MBA) `course-ai-digital-transformation`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `workflow, flow, table, compare, bars`

Architectural Theme: Enterprise AI Value Equations, Robotic Process Automation & AI Governance

> ***Pedagogical Whiteboard Model:*** *Illustrate enterprise AI value creation frameworks, RPA bot automation process flows, AI governance risk-level classifications, and digital maturity phase roadmaps.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Digital Transformation imperatives: Legacy modernization vs Digital-first paradigms**	Template `workflow` : Animated step-by-step state progression
Block 2	Days 6–10	**Enterprise AI Value Equation: Cost reduction, Revenue acceleration & Customer experience**	Template `flow` : Animated step-by-step state progression
Block 3	Days 11–15	**Robotic Process Automation (RPA): Workflow orchestration & Repetitive task bot automation**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Generative AI workplace deployment: Departmental copilot adoption & Prompt standards**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**AI Governance & Ethics: Bias auditing, Privacy protection, IP safety & EU AI Act compliance**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Organizational Change Management: Cultural readiness, Upskilling roadmaps & ROI dashboards**	Template `workflow` : Animated step-by-step state progression

Capstone Milestone: Enterprise AI Transformation Roadmap & Governance Charter: Multi-year digital transformation strategic roadmap with departmental RPA automation blueprints and ethical AI policies.

Course 32: Computer Literacy, Digital Productivity & OS Fundamentals `course-computer-fundamentals`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, boxes, tree-graph, table, compare`

Architectural Theme: Von Neumann Architecture, POSIX File Permissions, CLI Pipelines & TCP/IP Networking

> ***Pedagogical Whiteboard Model:*** *Diagram Input-Process-Output CPU bus flows, Unix command pipe data streams, file system inode permission trees, and TCP/IP 4-layer packet traversals.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Computer architecture: CPU registers, Arithmetic Logic Unit (ALU), RAM & Bus topology**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Operating Systems fundamentals: Kernel mode vs User mode, System calls & Process threads**	Template `boxes` : Animated step-by-step state progression
Block 3	Days 11–15	**POSIX Command Line: Terminal navigation, File permissions (chmod/chown) & Pipe redirection**	Template `tree-graph` : Animated step-by-step state progression
Block 4	Days 16–20	**Memory and storage hierarchy: L1/L2/L3 CPU Caches, Virtual memory paging & RAID arrays**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Computer networking basics: TCP/IP 4-layer model, IPv4 subnetting, DNS & Port routing**	Template `compare` : Animated step-by-step state progression
Block 6	Days 26–30	**Digital security & System health: Process monitoring, SSH keys & Preventive maintenance**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Systems Administration & Networking Verification Suite: Practical command-line and systems automation project demonstrating process monitoring, shell scripting, and network diagnostics.

Course 33: Everyday AI Literacy & Prompt Engineering `course-ai-prompt-literacy`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 21.2 min (18-25m Standard) | **Allowed:** `flow, compare, boxes, table, bars`

Architectural Theme: Prompt Engineering Frameworks, Chain-of-Thought Reasoning & Fact Verification

> ***Pedagogical Whiteboard Model:*** *Show prompt before/after optimization comparisons, Chain-of-Thought reasoning step paths, temperature sampling spaces, and fact-checking verification scorecards.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Generative AI mechanics: Probabilistic language modeling, Tokens & Prediction spaces**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Prompt structuring foundations: Context, Persona, Instruction & Constraint frameworks**	Template `compare` : Animated step-by-step state progression
Block 3	Days 11–15	**Advanced prompting techniques: Few-shot examples, Chain-of-Thought (CoT) & Decomposition**	Template `boxes` : Animated step-by-step state progression
Block 4	Days 16–20	**Multimodal AI capabilities: Text-to-image prompting, Audio transcription & Visual analysis**	Template `table` : Animated step-by-step state progression
Block 5	Days 21–25	**Workplace productivity automation: Research synthesis, Executive drafting & Data extraction**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Cognitive guardrails: Hallucination identification, Source grounding & Ethical evaluation**	Template `flow` : Animated step-by-step state progression

Capstone Milestone: Master Prompt Engineering Playbook: Production-grade collection of verified, structured prompt templates for workplace research, communication, data analysis, and validation.

Course 34: Excel & Data Analysis Fundamentals

`course-excel-data-viz`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `table, cells, bars, compare, flow`

Architectural Theme: 2D Coordinate Grids, Logical Evaluation, XLOOKUP Intersections & Executive Pivot Dashboards

> ***Pedagogical Whiteboard Model:*** *Illustrate 2D grid cell referencing, XLOOKUP array search traversals, Pivot Table multidimensional aggregations, and executive waterfall chart components.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Spreadsheet grid architecture: 2D coordinate referencing, Absolute ($) vs Relative locks**	Template `table` : Animated step-by-step state progression
Block 2	Days 6–10	**Statistical functions: SUM, AVERAGE, COUNT, MIN/MAX & Weighted average calculations**	Template `cells` : Animated step-by-step state progression
Block 3	Days 11–15	**Logical evaluations: Single IF, Nested IF statements & Multi-condition AND/OR formulas**	Template `bars` : Animated step-by-step state progression
Block 4	Days 16–20	**Text sanitization & Date calculations: TRIM, CLEAN, TEXTJOIN, CONCAT & DATEDIF math**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Lookup & Reference functions: Classic VLOOKUP vs Modern XLOOKUP & 2-Way INDEX-MATCH**	Template `flow` : Animated step-by-step state progression
Block 6	Days 26–30	**Data visualization & Reporting: Pivot Tables, Multi-dimensional slicing & Executive charts**	Template `table` : Animated step-by-step state progression

Capstone Milestone: Executive Financial & Operational Excel Model: Fully automated corporate spreadsheet workbook featuring dynamic XLOOKUP models, Pivot Table aggregations, and executive visual dashboards.

Course 35: Git, GitHub & Version Control Basics

`course-git-version-control`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `tree-graph, flow, states, compare, boxes`

Architectural Theme: Directed Acyclic Graphs, 3-Tier Git Areas, 3-Way Merges & Collaborative PRs

> ***Pedagogical Whiteboard Model:*** *Draw 3-tier Git architecture (Working directory -> Staging -> Repository), Directed Acyclic Graph (DAG) commit trees, and 3-way merge conflict marker resolutions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Version control architecture: Working Directory, Staging Area (Index) & Local Repository**	Template `tree-graph` : Animated step-by-step state progression
Block 2	Days 6–10	**Core versioning operations: Commit SHA hashes, Atomic commits, Git log & History inspection**	Template `flow` : Animated step-by-step state progression
Block 3	Days 11–15	**Branching dynamics: Pointer movements, Fast-forward merges & Feature branch isolation**	Template `states` : Animated step-by-step state progression
Block 4	Days 16–20	**Branch reconciliation: 3-way merge algorithms, Merge conflict resolution & Interactive rebase**	Template `compare` : Animated step-by-step state progression
Block 5	Days 21–25	**Remote collaboration: Git remotes, Push/Pull mechanics, Upstream tracking & Fetching**	Template `boxes` : Animated step-by-step state progression
Block 6	Days 26–30	**GitHub team workflows: Pull requests, Code review etiquettes, Branch protection & Tags**	Template `tree-graph` : Animated step-by-step state progression

Capstone Milestone: Multi-Branch Collaborative Repository Simulation: Realistic engineering project demonstrating trunk-based branching, clean rebase histories, resolved merge conflicts, and peer-reviewed PRs.

Course 36: Professional Tech Communication & Interview Mastery `course-softskills-communication`

Visuals: ⏳ Needed (0/30) | **Duration:** ⚠️ 6 min (Expansion to 18-25m Needed) | **Allowed:** `flow, compare, table, states`

Architectural Theme: Minto Pyramid Structured Messaging, Active Listening, STAR Method & Tech Interviews

> ***Pedagogical Whiteboard Model:*** *Diagram Minto Pyramid top-down communication hierarchies, STAR method (Situation, Task, Action, Result) storytelling flows, and email tone before/after comparison tables.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Foundations of technical communication: Clarity, Conciseness & The BLUF principle**	Template `flow` : Animated step-by-step state progression
Block 2	Days 6–10	**Structured thinking models: Minto Pyramid, MECE principle & Executive summaries**	Template `compare` : Animated step-by-step state progression
Block 3	Days 11–15	**Professional written correspondence: High-impact emails, Slack etiquette & Documentation**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Verbal presentations: Agile standup updates, Technical demos & Slide design principles**	Template `states` : Animated step-by-step state progression
Block 5	Days 21–25	**Interpersonal dynamics: Active listening, Constructive feedback loops & Conflict de-escalation**	Template `flow` : Animated step-by-step state progression
Block 6	Days 26–30	**Interview mastery: The STAR method, Behavioral question responses & Salary negotiation**	Template `compare` : Animated step-by-step state progression

Capstone Milestone: Executive Career Portfolio & Interview Masterclass: Comprehensive career communication asset package featuring structured technical project demos, STAR behavioral interview decks, and executive writing samples.

Course 37: Natural Language Processing & Computational Linguistics `course-nlp`

Visuals: ✅ Built (30/30) | **Duration:** 🎯 21.5 min (18-25m Standard) | **Allowed:** `letters, cells, table, flow, bars, tree-graph`

Architectural Theme: Tokenization, TF-IDF Vectors, Word2Vec Semantics & Transformer Self-Attention

> ***Pedagogical Whiteboard Model:*** *Show token sequence sliding windows, word embedding vector spaces with cosine distance, and Transformer Query-Key-Value self-attention matrix products.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	**Text preprocessing pipelines: Unicode normalization, Tokenization, Stemming & Lemmatization**	Template `letters` : Animated step-by-step state progression
Block 2	Days 6–10	**Statistical representation: Bag-of-Words, N-grams & Term Frequency-Inverse Document Frequency**	Template `cells` : Animated step-by-step state progression
Block 3	Days 11–15	**Distributed vector embeddings: Continuous Bag of Words (CBOW), Skip-gram & Word2Vec geometry**	Template `table` : Animated step-by-step state progression
Block 4	Days 16–20	**Sequence modeling: Recurrent Neural Networks (RNN), Hidden states & LSTM/GRU gating mechanisms**	Template `flow` : Animated step-by-step state progression
Block 5	Days 21–25	**Transformer architecture: Scaled dot-product Self-Attention, Multi-head matrices & Positional encodings**	Template `bars` : Animated step-by-step state progression
Block 6	Days 26–30	**Computational language applications: Named Entity Recognition (NER), Sentiment scoring & Summarization**	Template `tree-graph` : Animated step-by-step state progression

Capstone Milestone: Production Natural Language Processing Engine: End-to-end computational linguistic pipeline performing text preprocessing, TF-IDF/Dense embedding extraction, and sentiment/NER classification.

6. Phased Implementation Roadmap for the 24 Standalone Courses

Phase	Scope & Objectives	Verification Gate & Deliverables
Phase 1: Curriculum Expansion	Expand 23 legacy pilot courses into 6-part LongLessons (1,200–1,500 words per day) to hit the 18–25 min standard.	`estimateLessonMinutes` outputs 18.0 to 24.5 min across all 30 days.
Phase 2: Visual Suite Generation	Author visual specifications for the 24 courses across 144 blocks (720 days) using the 5-Step Loop.	Gate v2 (Rules R1–R14) outputs 100% PASS with deterministic AST bindings.
Phase 3: Student Simulation	Run `simulate-student-view.mts` across all 1,110 days and 6,660 lesson parts.	0 broken bindings, 0 render crashes, 100% clean full-matrix simulation.